TEXORA · LECTURE SHEET

Neps

A nep is a small knot of tangled fibres — millimetres across, yet able to condemn a whole fabric lot when it refuses the dye. This sheet defines neps, separates their types and causes, and sets out how they are counted and controlled.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 6 · LEVEL 3 TERM 1

1. What a nep is

A **nep** is a small, tight entanglement of fibres — typically 1 mm or less across, with a dense core — that behaves as a single particle rather than as individual fibres. In yarn it appears as a local thick speck; in dyed fabric, neps containing immature fibre dye paler than the ground and show as **white specks**, one of the most common causes of rejected dyed cotton goods.

Types of nep

- **Mechanical (processing) neps** — formed by the machinery itself: fibres rolled and tangled by card clothing, tight settings, damaged wires, excessive speeds or overloaded feed. They are mostly pure fibre tangles and can be reduced by correct machine settings and maintenance.
- **Biological (raw-material) neps** — present in the bale before processing: tangles built around immature and dead fibres, seed-coat fragments with fibres attached (**seed-coat neps**), and motes. Ginning creates many of them. They are largely a cotton-quality property (Topic 5: immature fibres, being flat, weak and lacking rigidity, tangle and wrap around any nucleus).

A practical distinction follows: mechanical neps are the mill's fault and yield to settings; biological neps arrive with the cotton and can only be reduced — by carding intensity, combing, and careful ginning upstream — never fully removed.

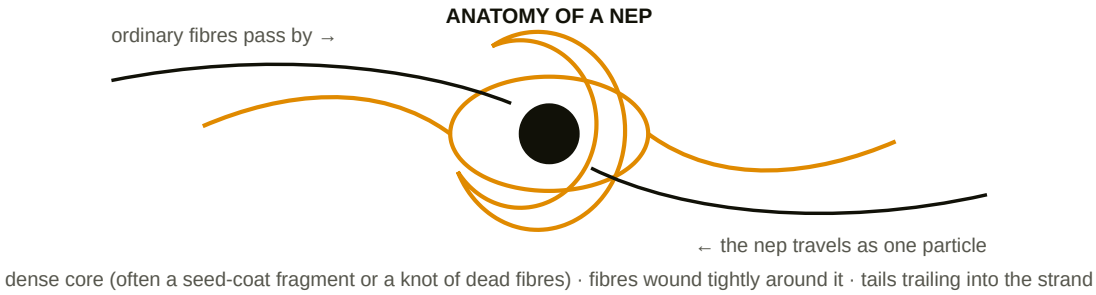


Figure 1 — Structure of a nep. Once wound tight, card wires can open only the loose outskirts; the core survives unless the nep is removed whole — which is what carding waste and comber noil accomplish.

2. Where neps are created and removed

STAGE	EFFECT ON NEP COUNT
Ginning	Creates neps — harsh or damp ginning is a primary source; saw-gin more than roller-gin.
Opening & cleaning	Creates some (beating action) while removing trash; aggressive opening of immature cotton raises neps.
Carding	The great nep *remover* of the carded route: a well-set card typically removes the majority of incoming neps (nep removal efficiency commonly 60–80 %+ across the card), while a badly set or overloaded card creates them.
Combing	Removes most remaining neps with the noil — the reason combed yarns dye cleanest.
Drawing, roving, spinning	Should add few; worn aprons/cots and poor roller settings roll fibres into new neps.

3. Measurement of neps

3.1 Manual / template counting

A web or a weighed tuft of fibres is spread thinly over a contrasting (black) board and neps visible against the background are counted, sometimes against a standard template defining minimum nep size. Results are expressed per gram or per unit area of web. Slow and operator-dependent, but it needs no instrument and remains a useful mill-floor check on card web.

3.2 AFIS nep measurement (routine reference)

AFIS (Topic 9) separates a fibre sample into individual fibres and counts nep particles electro-optically, reporting **nep count per gram** and **mean nep size (µm)**, distinguishing fibre neps from seed-coat neps.

Typical orders of magnitude: raw cotton may carry 200–400 neps/g; a good card sliver should be down to roughly 20–60 neps/g; combed sliver lower still. Exact expectations depend on the cotton — judge a machine by its *change* in nep count (removal efficiency), not by an absolute number copied from another mill.

$$\text{Nep removal efficiency \%} = [(\text{neps in} - \text{neps out}) / \text{neps in}] \times 100$$

WORKED EXAMPLE — COUNTING AND CARD EFFICIENCY

(a) A 0.50 g sample of card *feed* material, spread and counted in five fields, contains 18, 22, 19, 24 and 17 neps — total 100 neps → 100 / 0.50 = **200 neps per gram**. A figure of this size is normal for feed material; the same figure in delivered sliver would condemn the card.

(b) Card feed measures 280 neps/g and card sliver 56 neps/g. Efficiency = (280 − 56)/280 × 100 = **80 %** — a healthy card. If next month the same cotton gives 40 % efficiency, the cotton has not changed: the card clothing, settings or feed rate have.

3.3 Nep counting in yarn

Yarn neps are counted as an imperfection class by evenness testers (Topic 13: neps at the +200 % mass threshold per kilometre) and appear again as hairiness-adjacent faults in fabric inspection. Keep the three populations distinct in reports: fibre neps per gram (AFIS), card-web neps (visual) and yarn neps per km (evenness tester) are not interchangeable numbers.

4. Causes checklist — when nep count rises

- **Raw material:** high immature/dead fibre content (low maturity ratio, Topic 5); harsh ginning; damp cotton at ginning or opening.
- **Card:** dull or damaged cylinder/doffer clothing; cylinder–doffer and cylinder–flat settings too wide (neps pass unopened) or flats overloaded; feed too heavy for the cotton; excessive production rate.
- **Preparation:** over-beating in opening lines; re-processing of waste with high nep content back into the mix.
- **Later machines:** damaged aprons and cots, eccentric rollers — rolling fibres instead of drafting them.

5. Control strategy

1. Buy and blend on maturity as well as micronaire; segregate high-nep lots to combed or undyed end uses.
2. Gentle opening; no more beating than the trash level requires.

3. Hold card condition: clothing sharpness, settings to the standard for the cotton, feed and speed within rating; monitor nep removal efficiency as a routine KPI.
4. Comb where the product cannot tolerate neps (fine dyed goods).
5. Test at the sliver, not only at the bale — the process trend finds the faulty machine while there is still time to fix it.

Glossary

Nep — a small tight entanglement of fibres behaving as one particle. **Seed-coat nep** — a nep built around a fragment of seed coat with attached fibres. **Mechanical nep** — a nep created by processing machinery. **Biological nep** — a nep present in the raw material. **Nep removal efficiency** — percentage reduction in nep count across a machine. **White speck** — pale-dyeing nep visible in dyed fabric. **Mote** — an aborted seed / seed fragment found in cotton.

Key points

• Neps = tangled fibre knots; immature fibres are their favourite building material.
• Carding is the main removal stage (target high removal efficiency); combing polishes what carding leaves.
• Report the population and basis: AFIS neps/g and size, web counts, or yarn neps/km — never mix them.
• A rising nep trend on unchanged cotton indicts machine condition, not the bale.
• Dyed-fabric white specks are yesterday's immature-fibre neps made visible.

Practice

1. A 1.00 g sample of sliver contains 42 neps counted by AFIS. The card feed contained 210 neps/g. Find the sliver nep count per gram and the card's nep removal efficiency.
2. Classify: (a) nep formed around a seed-coat fragment in the bale; (b) nep appearing after a card's clothing dulled. State the remedy class for each.
3. Why does a low-maturity cotton nep more in the card even when the card is perfectly set?
4. A mill reports "neps = 35" with no other information. List four things the report must add before the figure can be used.

Answer key

1. Sliver = **42 neps/g**. Efficiency = (210 − 42)/210 × 100 = **80 %**.

2. (a) Biological / seed-coat nep — remedy lies in raw-material selection, ginning quality and removal by carding/combing. (b) Mechanical nep — remedy lies in machine maintenance: grind/replace clothing and reset the card.
3. Immature fibres are flat, weak and limp: they buckle, wrap around any nucleus and roll into tangles instead of being opened and separated by the card wires, and they break into short fragments that re-tangle. Perfect settings minimise, but cannot abolish, a raw-material tendency.
4. The material and stage (raw cotton / sliver / yarn), the method (AFIS, visual web count, evenness tester), the basis (per gram, per unit area, per km) with threshold if a yarn figure, and the sample mass or length tested (with the cotton's identity/maturity for context).

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